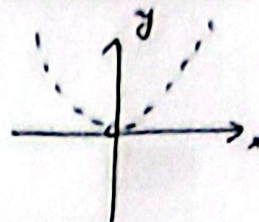


Exer 10P

$$1) F(x, y) = \frac{y-x}{x^2 y} \quad DF = \{ (x, y) \in \mathbb{R}^2 / y \neq x^2 \}$$



$$C_{-1}(F) = \{ (x, y) \in DF / y - x = y - x^2 \}$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

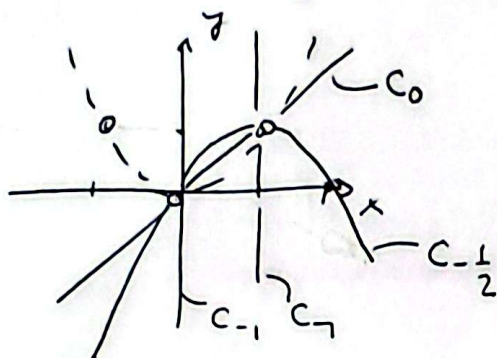
$$\begin{cases} x=0 \\ x=1 \end{cases} \quad \begin{array}{l} \text{dos} \\ \text{Rectas} \\ \text{que se excluyen} \end{array}$$

$$C_{-\frac{1}{2}}(F) = \{ (x, y) \in DF / y - x = \frac{1}{2} y - \frac{1}{2} x^2 \}$$

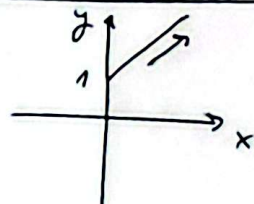
$$\hookrightarrow \frac{1}{2} y = -\frac{1}{2} x^2 + x \rightarrow y = -x^2 + 2x$$

parábola q' p'tos excluidos

$$C_0(F) = \{ (x, y) \in DF / y = x \} \quad \text{Recta sin el (0,0)}$$



$$2) \sigma(t) = (t^2, t^2+1). \quad \begin{cases} x=t^2 \geq 0 \\ y=t^2+1 \end{cases} \rightarrow y = x+1, x \geq 0.$$



$$\text{busco } t / \sigma(t) = (4,5) \quad t > 0 \quad |t|=2 \quad t=2 \vee t=-2 \notin D\sigma$$

$$\sigma'(t) = (2t, 2t) \rightarrow \sigma'(2) = (4,4) \quad \|\sigma'(2)\| = \sqrt{32} = 4\sqrt{2}$$

$\hookrightarrow$ velocidad $\hookrightarrow$ rapidez

$$3) a) \nabla F(1,1) = \left(y \cos(x-1) + e^{x-y} + 2y^2 - \frac{1}{x}, \sin(x+1) - e^{x-y} + 4xy - \frac{1}{y} \right) \Big|_{(1,1)} =$$

$$\nabla F(1,1) = (3,2). \quad \|\nabla F(1,1)\| = \sqrt{13} \rightarrow \text{valor de } d\text{d máx}$$

$$-\sqrt{13} \text{ valor de } d\text{d mín} \quad \left(\frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}} \right) \text{ dirección de } d\text{d máx}$$

$$\left(-\frac{3}{\sqrt{13}}, -\frac{2}{\sqrt{13}} \right) \text{ dirección de } d\text{d mín.}$$

$$b) 9 < 13 < 16 \Rightarrow 3 < \sqrt{13} < 4$$

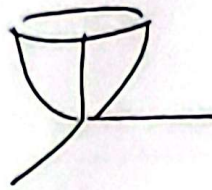
$\hookrightarrow$ como $\sqrt{13}$ es el valor máximo

no hay ninguna dirección de la cual

la derivada direccional sea igual a 4. (Pues $4 > \sqrt{13}$).

4) $F(x, y) = x^2 + y^2 \quad DF = \mathbb{R}^2$

a) $z = x^2 + y^2$, parabolóide



$S = \text{Im}(\tau)$

propriedade

$\tau: \mathbb{R}^2 \rightarrow \mathbb{R}^3 / \tau(u, v) = (u, v, u^2 + v^2)$

outra propriedade

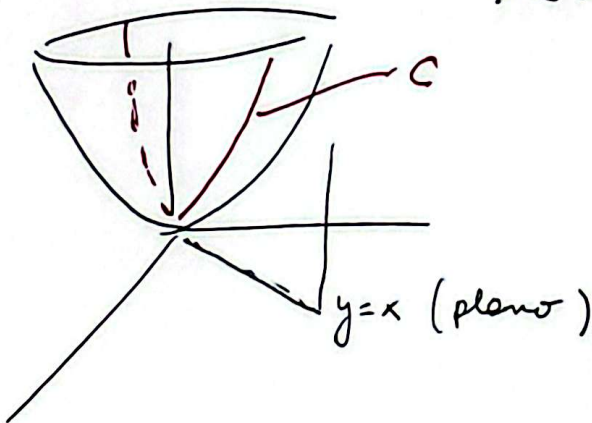
$\tau^*: D \rightarrow \mathbb{R}^3 / \tau(u, v) = (u \cos v, u \sin v, u^2)$

com $D = \{(u, v) \in \mathbb{R}^2 / u \geq 0, v \in [0, 2\pi]\} = [0, +\infty) \times [0, 2\pi]$

b) $C: \begin{cases} z = x^2 + y^2 \\ y = x \end{cases} \Rightarrow z = 2x^2$

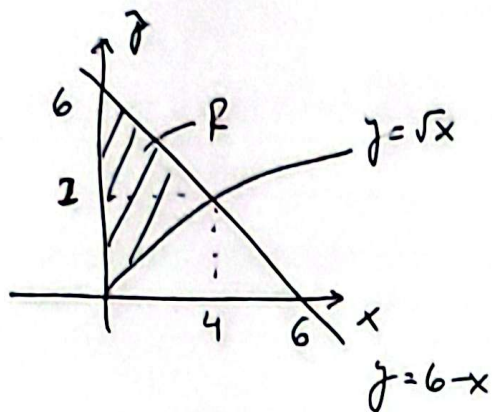
propriedade σ ; $C = \text{Im}(\sigma)$ sendo $\sigma: D \subseteq \mathbb{R} \rightarrow \mathbb{R}^3 /$

$\sigma(t) = (t, t, 2t^2) \quad D = \mathbb{R}$



C es una parábola en $\mathbb{R}^3$.

1)



$$\sqrt{x} = 6 - x$$

$$x = (6 - x)^2$$

$$x = 36 - 12x + x^2$$

$$x^2 - 13x + 36 = 0$$

$$\frac{13 \pm \sqrt{13^2 - 4 \cdot 36}}{2} = \frac{13 \pm 5}{2} \begin{cases} x_1 = 9 \\ x_2 = 4 \end{cases}$$

$$a(R) = \int_0^4 \int_{\sqrt{x}}^{6-x} dy dx = \int_0^2 \int_0^{y^2} dx dy + \int_2^6 \int_0^{6-y} dx dy.$$

$$\int_0^4 (6-x - \sqrt{x}) dx = 6x - \frac{x^2}{2} - \frac{2}{3} x^{3/2} \Big|_0^4 = 24 - 8 - \frac{16}{3} = \frac{32}{3}$$

b) Como la circulación es sobre el borde de R (curva cerrada) uso Green. Recorriendo la curva en sentido horario.

$$\oint_{\partial R} F \cdot d\vec{s} = \iint_R (Q_x - P_y) dA = \iint_R 4 dA = 4 a(R) = 4 \cdot \frac{32}{3}$$



$$2) a) JF(x,y) = \begin{pmatrix} \frac{y}{xy+1} + ye^{xy} & \frac{x}{xy+1} + xe^{xy} \\ -y \sin(xy) & -x \sin(xy) \end{pmatrix}$$

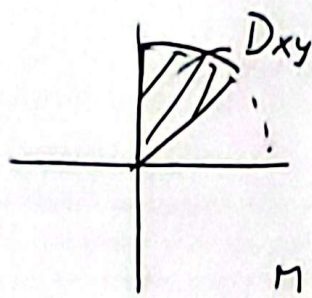
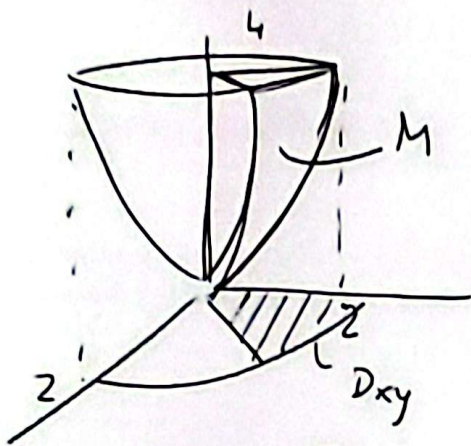
$$J\vec{g}(t) = \begin{pmatrix} e^t \cos t \\ -\sin t - 2e^{2t} \end{pmatrix} \quad \vec{g}(0) = (1, 0)$$

$$b) \sigma'(0) = JF(1,0) \cdot J\vec{g}(0) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \vec{n} \text{ (normal)}$$

$$\|\vec{n}\| = 2$$

↳ Toroid.

3)

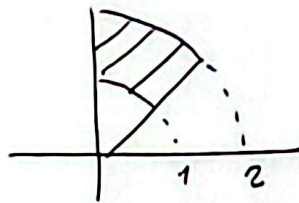


$$\begin{cases} x = r \cos \theta & 0 \leq r \leq 2 \\ y = r \sin \theta & \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2} \end{cases}$$

$$M = \begin{cases} x = r \cos \theta & 0 \leq r \leq 2 \\ y = r \sin \theta & \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2} \\ z = 2 & r^2 \leq z \leq 4 \end{cases}$$

$$\begin{aligned} a) \quad V(M) &= \int_0^2 \int_{\pi/4}^{\pi/2} r(4-r^2) d\theta = \int_{\pi/4}^{\pi/2} \left(4r^2 - \frac{r^4}{4} \right) \bigg|_0^2 d\theta = \\ &= \frac{\pi}{4} \cdot [2 \cdot 4 - 4] = \frac{\pi}{4} \cdot 4 = \pi. \end{aligned}$$

$$b) \quad D_{xy} \text{ con } x^2 + y^2 \geq 1$$



$$\begin{aligned} a(r) &= \int_0^2 \int_{\pi/4}^{\pi/2} r d\theta dr = \\ &= \frac{\pi}{4} \cdot \frac{r^2}{2} \bigg|_1^2 = \frac{\pi}{4} \left[2 - \frac{1}{2} \right] = \frac{\pi}{4} \cdot \frac{3}{2} = \frac{3\pi}{8} \end{aligned}$$

$$4) \quad C: \begin{cases} y = 4 - x^2 \\ 0 \leq x \leq 2 \end{cases}$$

$$\sigma: [0, 2] \rightarrow \mathbb{R}^2 / \begin{cases} \sigma(t) = (t, 4 - t^2) \\ \sigma'(t) = (1, -2t) \end{cases}$$

$$a) \quad m(C) = \int_0^2 2t \sqrt{1+4t^2} dt =$$

$$\|\sigma'(t)\| = \sqrt{1+4t^2}$$

$$2 \int_0^2 t \sqrt{1+4t^2} dt = \frac{1}{4} \int_1^5 \sqrt{u} du = \frac{1}{4} \cdot \frac{2}{3} u^{3/2}$$

$$\textcircled{5} \begin{cases} u = 1+4t^2 \\ du = 8t dt \quad \frac{1}{8} du = t dt \end{cases}$$

$$= \frac{1}{6} \cdot \frac{2}{3} (1+4t^2)^{3/2} \bigg|_0^2 = \frac{1}{6} [17\sqrt{17} - 1].$$

$$b) \quad F \text{ es conservador!}$$



Para no aplicar T. Green (+ cuentas)
hallamos ϕ tal que $\nabla \phi = F$.

$$\phi = e^{x+y} + 2xy^2 - e^x y \Rightarrow \int_C F ds = \int_{(0,4)}^{(2,0)} F ds = \phi(2,0) - \phi(0,4) = e^2 - e^4 - 4$$